Data preprocessing

- it refers to the techniques and procedures used to prepare raw data into a clean, organised and structured format suitable for analysis or modling

→ improved data Quality

→ Enhancing model performance

→ Reducing computational complexity

→ ensuring compatibility

various techniques :-

① Handling missing values

→ Imputation, deletion or prediction-based method.

② Handling outliers :-

→ Trimming or transformation-based method.

— : Data types and forms : —

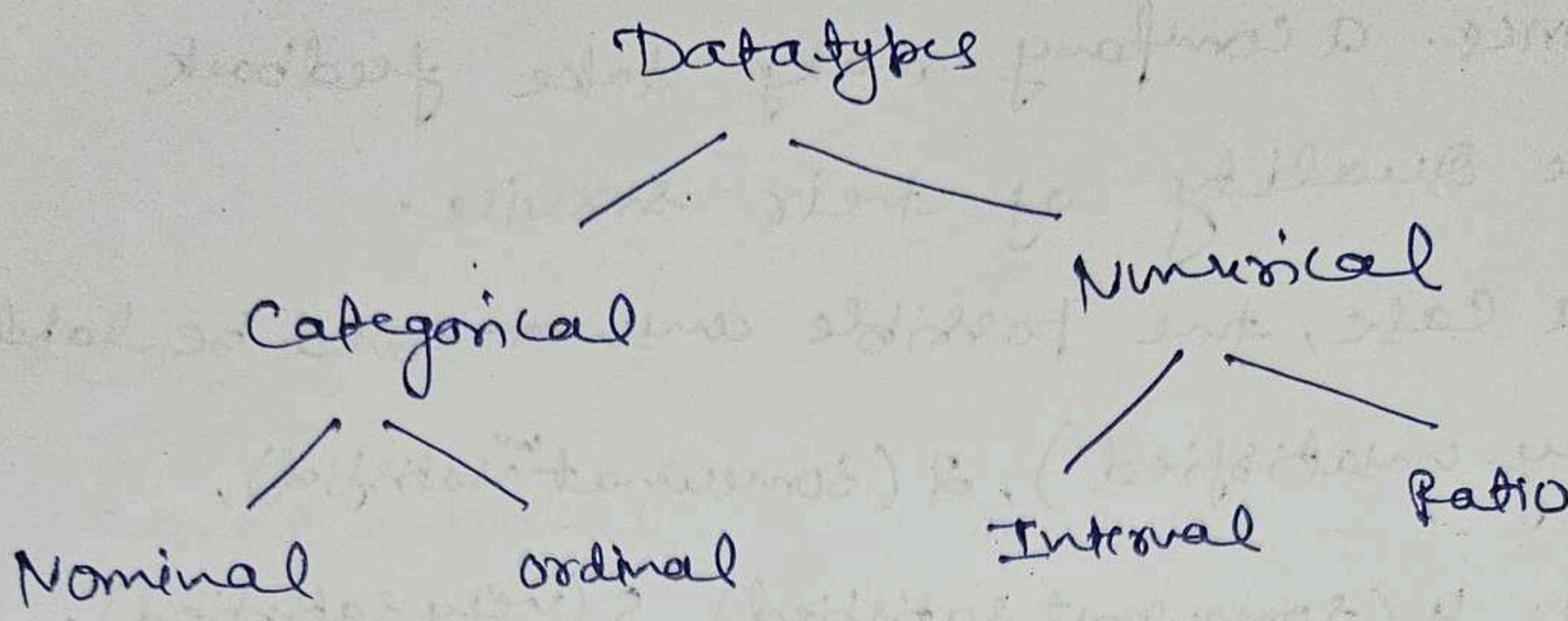
(3) Normalization / scaling :- min-max scaling, Z-score normalization, or robust scaling

(4) encoding categorical variables : - one hot encoding, label encoding or target encoding.

(5) feature extraction :- selecting relevant features or transforming existing features.

(6) Data splitting :- splitting data into train, validation and test sets for model evaluation.

— ! Data types and forms ! —



Categorical data :-

This type of data is non-numeric and consist of text that can be coded as numeric. However these Nubers do not represent any fixed mathematical notion or meaing for the text and they are simply assigned as labels or codes.

① Nominal data :- this type of data is used to label variables without providing any quantitative value.

. Nominal scals are mutually exclusive that is, the valus don't overlap.

· ordinal data :- this type of data is where label variables that need to follow some order for instance. a company may take feedback about the quality of their service.
in such a case, the possible answer could be labled as 1 (very unsatisfied), 2 (somewhat unsatisfied), 3 (neutral), 4 (somewhat satisfied), 5 (very satisfied).
thus, each categorical value is assigned on a rating scale of 1 to 5.
· ordinal data follows some order of preference, satisfaction, comfort, happiness, or any such similar order.

② Numerical data :- this type of data is numeric and it usually follows an order of values. these quantitative data represent fixed values.

③ Interval data :-

→ equal interval b/w value
→ No True zero value (zero mean not Absolute zero)
Exp temperature like (0°C) Time

clothe of new born baby size.
(zero size)

Ratio Scale

→ equal interval.

→ Absolute zero exist ,

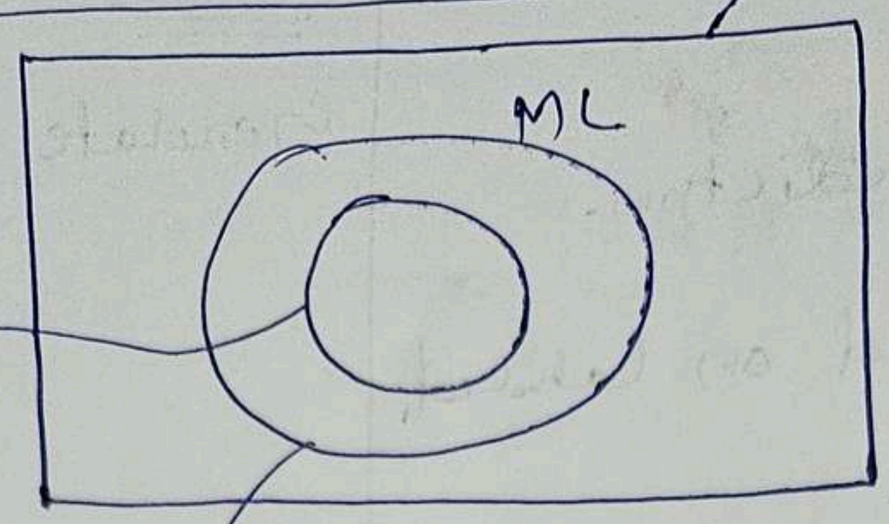
Exp Height, weight, income,

Conclusion :-

	Nominal	ordinal	Interval	Ratio
Naming	✓	✓	✓	✓
ordering		✓	✓	✓
equal interval			✓	✓
true zero				✓

AI VS ML VS DL

AI - Build an Application, that can perform its own task with out human intervention.



Ex:
1. Netflix → Recommendation system.
2. Self driving car.
3. Robotics Navigation
4. Face Recognition,
5. sentimat analysis

given product = data

Mimic Human brain

ML is a subset of AI

ML → Stats tools to perform statistical analysis, visulization, prediction and forecasting

understad about the data.

Deep learning (three important things we know learn)

- ANN
- CNN and object detection. → computer vision.
- RNN and its variants → Text Related. usecases
  → Time series usecases.

⇓

{ Transform & BERT }

⇓

Generative AI

Two types of models used in ~~industry~~ using.

Models

- Discriminative
- Generative AI

Data scien project

Tasks

classification, prediction.

Dataset:- Trained on labeled Dataset.

Tasks

Generate new data a

— ! LLM Models ! —

OpenAI, Meta, Google, Anthropic.

GPT4 → GPT5

LLama2 → LLama3

Gemini → Gemma

Claude3

Best

Foundation models

pretrained models.

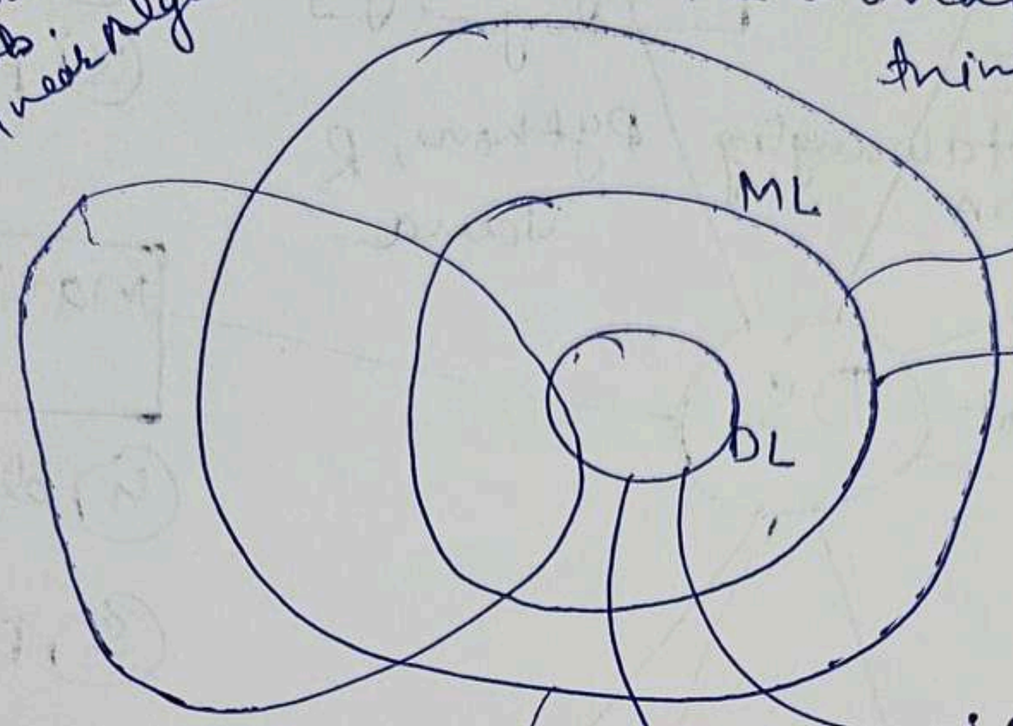
Huge data

fine tuning

Domain Specific usecase

Data sciem project lifecycle

AI VS ML VS DL VS DS



AI - enables the machine to think; without human intervention.
Mixed of two approaches
1. supervised 2. unsupervised.
3. Reinforcement learning.
(semi supervised)
provides statistical tools to explore the data.

MultiNeuralN/w Architecture

Can we make a machine to learn mimic of human brain.

ANN, CNN, RNN
TL. (Transfer learning

How to learn Data Science:—

DS

① Data Analytics
- Datavisualization
  1. TABLEAU
  2. Power BI
  3. Matplotlib, seaborn, chiplot

② Programming Language
- FE
- Data wrangling
- EDA
- Python, R, Java

③ Maths
1. Statistics
2. Linear Algebra
3. Differential calculus

④ Machine learning
1. Classification
2. Regression
3. Reinforcement learning
4. deeplearning
5. Dimensionality Reduction
6. Clustering algorithm

⑤ Web scrapping
1. Beautiful soup.
2. Scrapy.
3. URLLIB.

⑥ IDE — integrated development environment.
1. Pycharm
2. Jupyter
3. Spyder

⑦ Deployment
AWS, Azure,

II ult

Sources of data

① Primary.

— surveys

— interviews

— Experiments

— Observations

② Secondary data sources

— Government databases

— Research publications

— Open databases. (Kaggle, UCI Repository)

— website and blogs

③ web and social media

— Twitter, Facebook, Instagram

— News websites

— E-commerce platforms

④ Sensors and IoT devices

GPS, temperature, traffic sensors

Smart watches

unit III The process of inspecting, cleaning, transforming and modeling data to discover useful information, pattern, and insights.

EDA

unit IV

Data visualization is the graphical representation of information and data using visual elements like charts, graphs, maps and plots.

Various visulization Techniques:-

1. Bar chart
2. Line chart
3. Histogram
4. Scatter plot
5. Boxplot
6. pie chart
7. Heat map.

Unit V Recent Trends in data science

(1) Healthcare : predictive diagnosis, disease risk prediction, medical image analysis

(2) Finance : fraud detection, credit scoring, algorithm trading

(3) Retail & Ecommerce : Recommendation system, customer segmentation, inventory management

(4) social media : sentiment analysis, user behavior predictions, targeted advertising

(5) manufacturing : predictive maintenance, supply chain optimizations

(6) Agriculture : Crop yield predictions, soil analysis, weather impact modeling.

(7) Smart cities : Traffic predictions, energy

troduction to Data science :-

ata science is an interdisciplinary field that ses mathematics, statistics, computer sciens and domain knowledge to extract meaningful insights and knowledge from structured and unstructured data.

Why data sciens is important :-

① Helps in better decision-making

Data sciens provides data driven intigrits that support strategic busines and opelational decisions

Exp. A bank uses customer transaction data to decide whether to approve a loan.

② Unlocks Busines Intigrits

Data sciens helps companies discover hidden patterns and trends from raw data.

Exp. Analyzing online shopping behavior to recommend relevant products.

(3) Enhances predictive power

Machine learning models in data science can predict future event with high accuracy.

Exp predicting whether a patient is likely to develop diabetes based on health records.

(4) Support Automation

Automated systems powered by data science reduce manual effort and increase efficiency.

Exp:- Fraud detection systems that automatically flag suspicious credit card transactions

(5) Improves customer Experience

personalized recommendation and targeted marketing improve user satisfaction.

Exp. Netflix recommending shows based on your viewing history.

Real time decision making
tools like real time dashboards, companies
can act quickly based on fresh data.

Exp stock market algorithms that responds
instantly to price changes.

(7) Gives Competitive Advantage! -
organizations that adopt data sciew
stay ahead. of competitions by making smarter
and ~~deci~~ faster decisions.

Exp Uber uses real time data for route
optimization, pricing and demand prediction.

(8) Reduces Risk: -
Data sciew helps identify potential risk
and develop strategies to mitigate them.

Exp Insurance companies use predictive
models to asses customer risk profiles.

→ many more exp of the importance of data science :—

(1) Health care

problem : early detection of disease

(2) Education.

problem :- student dropout prediction

(3) Agriculture

problem :— Crop failure due to weather or pests.

Data science uses

→ using satellite and sensor data to monitor crop health

→ Forecasting rainfall and temperature trends

(4) Sports

problem : optimize team performance.

data science use :—

Tracking player statistics, injury history,

(5) Banking & finan:-

problem: Loan default or fraud detection.

Data sciem:-

using credit history, income and transaction, behavior.

→ Detecting suspicious patten automatically.

(6) E-commerce:-

problem:- Increase sales through personalization

Data sciem use:

- Analyzing user clickstreams, purchase history, and reviews.
- Recommending product and dynamic pricing.

(7) Transport and Logistics

Problem: Route optimization.

Data sciences:-

- GPS and traffic data used to optimize delivery routes.
- predict full. usage and maintenan schedul

(8) Manufacturing :-

problem : predict machine failure.

Data sciem use:

using IOT sensor data from machines.

(9) social media :--

problem: sentiment analysis and content filtering

(10) Retail :-

problem: Inventory management and customer demand :-

Data sciem use :-

predicting which item will be in demand, festival or seasons.

cyber security:-

problem: Real time threat detection.

data scientist use:-

Analyzing network traffic and login behavior.

(12) Energy sector

problem:- predict power consumption.

data scientist use

Analyzing temperature data, time and usage data.

(13) Entertainment:-

problem: Content recommendation

data scientist use:

- understanding viewer preferences from viewing history.
- Recommending song, videos or new articles.

(14) Smart cities

problem Traffic and waste management.

These exples show how data scientists

(1) saves time.

(2) saves money.

(3) Improves decision-making.

(4) provides better services

(5) solves real problems in every industry.

minologies related to data scienu:-

m	Explanation	Example.
taset	Collection of data	A CSV file of students marks.
eature	Independent variable or input	Age, income in a customer dataset
Label/Target	Defendent variable or output	whether a customer buys or not (Yes/No).
Training Data	Data used to train the model	80% of the dataset.
Testing data	Data used to evaluate the model.	Remaining 20%
Model.	The algorithm applied to data	Decision Tree, Linear Regression
Overfitting	model perform well on training data but poorly on testing data.	Too complex model memorizing data.

underfitting — model is too simple and misses patterns — lean m... & nonlin... pat... da...

Type of data :-

Type	Descriptions	Example
structured data	Tabular format (rows & columns)	Excel sheet, SQL tables
unstructured data	No predefined format	Emails, social meadia texts, images
semi-structured data	partially organized.	XML, JSON files.

Based on measurement scale :-

A measurement scale is a system used to categorize and quatify variables.
It defines the type of data, you are working

determines what kinds of mathematical statistical operations can be performed on that data.

Understanding the measurement scale of variables helps you.

- Choose the right statical method.
- Decide which graph or chart to use.
- Know whether you can calculate the mean, median, or mode.
- Choose a suitable machine learning algorithm.

Datatypes

Categorical — Nominal, ordinal

Numerical — interval, Ratio

① Nominal scale :→

Data is categorized into distinct groups without any order or ranking.

only names/labels are used to identify data.

- No mathematical operations (like addition comparison) are meaningful.

characteristics :-

- No ordering
- Cannot calculate mean or median
- only mode make senses.

Exp.

Feature	values
Gender	Male, Female
Blood type	A, B, AB, O
Color	Red, Green, Blue
Country	India, USA, Canada.

dinal Scale :-

ata is categorized with a meaningful
oder, but intervals b/w values are not
niform or meaningful.

- you can rank the data, but can't measure
the exact diffrens b/w values.

Characteristics:-

- ordered categories
- Diffrens b/w valu are not quantifiable.
- median and mode are meaningful, mean
is not

Exp	
Feature	ordered values
customer satisfaction	poor < fair < good < excellent
Education level.	Highschool < Bachelor < Master < PhD.
clothing size	small < medium < large < XL

Valid visulization :-

- Bar chart
- Box plot

(3) Interval scale :-

Numerical data with equal intervals b/w values.

characteristics

- ordered
- equal intervals
- can compute mean and standard deviation.

| इसके अन्दर Absolute zero exist नही करता है,
| zero की value arbitrary होती है.

Exp

Feature	values
age	1, 2, 3, 4, 5 (years)
income	0–20000, 20001 – 40000, 40001 – 60,000
temperature	10°C, 20°C, 30°C

contin:-

Internal scale

→ internal scale is a numerical scale

Ques:-

The difference (interval) b/w any two values is meaningful and consistent.

- But it has no true zero point (zero does not mean "none").

- therefore, multiplications and division, do not make sense.

-No true zero point:-

"NO true zero" means that zero point on the scale does not represent the complete absence of units being measured.

in an interval scale 0 does not mean absence of the quantity.

Exp

. 0°C does not mean no heat or no temperature.

. It just the freezing point of water, chosen arbitrarily.

so this is not a true zero

Exp (2)

IQ scores

— A score of 0 does not mean "no intelligence".

Exp (3) Calendar year

— year 0 does not mean "no time".

(4) Ratio scale :-

A Ratio scale is a numerical scale where:
- values have order and equal intervals.
- it has a true zero point (zero means "none" of the quantity).
- meaningful ratios can be calculated.

Exp weight

0 kg means "no weight"
20 kg is weight

- Height (cms or inches)
- Age (in years)

- income (in dollar)

$0 means "no income".

$100k is twice as much as $50k.

Distance

- all statistical operations are valid.

(addition, subtraction, multiplication, division)

Data scienu project lifecycle

Each data scienu project follows a logical and organized series of steps. Aese help transform raw data into meaningful insight and deployable solutions.

1. Problem Identification (understanding the problem)

understand the business problem clearly and convert it into a data scienu problem.

Activities

→ Discuss the stakholders

- identify the goal (prediction / classification / optimization)
- define success criteria.

Exp A bank wants to predict whether a customer will default on a loan.

→ Convert this into a binary classification problem

2. Data collection :-

Gather relevant and high-quality data from all possible sources.

- Collect data from databases, APIs, sensors, web scrapping, surveys.
- ensure sufficient volume and variety.

Sources

→ CSV file, SQL databases, cloud platforms, IOT sensors.

(3) Data preparation (Data cleaning & preprocessing)

Convert raw data into a usable form.

Activities :-

- Handling missing values (fill, drop)
- Remove duplicates
- Normalize / standardize data
- Encode categorial variable.

Detect and handle outliers.

P) Exploration (Exploratory data analysis - EDA)
understand data patterns, correlations and distributions.

Activities

→ summary statistics (mean, median, standard deviation)

→ visulization: histograms, boxplots, pairplots heatmaps.

→ identify relationship b/w feature and the target.

Tools

Python (Pandas, matplotlib, seaborn), R, excel.

Modeling (Building)

use algorithm to build predictive or descriptive model

Activities:-

→ select suitable algorithm (Logistic Regression, decision Trees).

→ Train model on training dataset.

&

→ Tune hyperparameters (using Grid search, cross validation)

6

→ Evaluation. :-

Measure how well the model performs using test data.

Activities

→ use performance metrics:

- classification: Accuracy, precision, Recall, F-1 Score, AUC.
- Regression: RMSE, MAE, R2
- confusion matrix analysis.

Models

describe

void overfitting / underfitting.

) Deployment

ntegrate the model into the real-world

Pegrovisonment for actual use:

Activities

use platform like flask, FastAPI, or Docker

(8) monitoring (post-deployment monitoring

and maintem.

- ensure the model continues to perform

well over time.

Activities :-

- Track model performam regularly.
- monitor data drift (changes in data patterns).
- updates/retrain model when accuray

drops.

Real-world Application of data scien

Domain	Application
Healthcare	Disease prediction.
Finance	fraud detection.
Retail	Recommendation engine
Transport	Traffic forecasting
Marketing	Customer segmentation.
Social media	Sentiment Analysis
Agriculture	Yield prediction / Forecast crop output data using weathe data.

Why data sciem is important

① Better decision making
Helps business make strategic decisions using data rather than guesswork.

② Competitive Advantages
Companies using data sciem can stay ahead in the market.

Scientific automation

Replaces manual process with intelligent automation (e.g. chatbots, fraud detection systems.

Datascience Toolkit :-

To perform data sciem efficiently, various tools and libraries are used.

Category	Tools/Library	description
Programming	Python, R	Most used language for DS.
Data handling	Pandas, Numpy	Data manipulation and numerical computing.
visulization	matplotlib, seaborn, plotly	create graphs and plots
Machine learig	Scikit-learn, Tensflow, Pytorch	Model building
Bigdata tools	Hadoop, Spark	For large datasets ~~Data Storage & querying~~

Database	MySQL, MongDB	Data storage and Query
IDEs	Jupyter, Google Colab, Rstudio	Development environments

What is Data Analysis: —

Data Analysis is the process of (Analyzing) inspecting, cleaning, transforming and modeling data to discover useful information, draw conclusions and support decision making.

There are four main types of data analysis: —

(1) Descriptive Analysis
(2) Diagnostic Analysis
(3) Predictive Analysis
(4) Prescriptive Analysis.

(1) Descriptive Analysis: —

Goal: To understand what happened in the past.

Step

1. Collect data — Gather past or pres
data (sales, attendan

2. Summarize data — use statistics like mean, median, modes.

3. Visulize results — charts, graphs, tables.

Exp

① A school summarize student scores
Average = 75, Higest = 98, Lowest = 45.

② A company ~~materials~~ reports that sales increased by 10% last month.

Tools

Excel (Sum, Average), Python (Pandas), R,
Tableau, Power BI.

Diagnostic Analysis :-

Goal: To understand "why something happened.

Step	Explanation.
1. Identify causal event	for Exp. sudden drop in attendance.
2. Compare patterns	Compare with other months, department.
3. Drill into data	use filters, segmentation, Correlations.

Exp: sales dropped by 25% in march.

- Diagnostic Analysis Issues:
  - → Competitor launched a new product.
  - → Customer complains increased.

- Tools

Ex: SQL, Python (seaborn, matplotlib), Power BI.

(3.) Predictive Analysis

Goal :- To forecast what is likely to happen in the future.

Step	explanation
1. Collect historical data.	Sales of past 12 months.
2. Train model.	use algorithm (e.g. linear Regression)
3. Predict outcome.	Estimate next month's sales, temp.

Exp: →

- A student with low attendance has a 70% chance of failing.
- Rain is likely tomorrow based on whether patterns.
- sales will rise by 15%. in next quarter

ols:

Machine learig (scikit-learn, Tensorflow)

Time series, Python, R.

4 Prescriptive Analysis:-

Goal:- To suggest what action to take.

Step	Explaination
1. We prediction data	from predictive models.
2. Add rules / Optimizations	Add constraints,
3. Recommend decision	suggest best possible action.

Exp

1. Based on predicted traffic, Google maps suggest the fastest route.
2. Amazon recommends the best product bundle to increase sales.
3. A School system suggest remedial classes for at risk studa

(Real life exp. (All 4 together):-

Scenario: A company reviews its prod. sales.

① Descriptive: Sales in June were 50,000 Rs.

② Diagnostic: Drop due to new competitior

③ Predictive: July sales projected at 60,000 Rs.

④ prescriptive: offer 10% discount to increase market share.